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Logic

- Systematic method of reasoning that helps to analyze statements and deductions based on their truth or falsity.
- E.g.

Today is Sunday → Simple mathematical statement

$n \geq 0$, $1+2+3+\dots+n = \frac{(n)(n+1)}{2}$ → logical statement

↳ can be proven using induction.

Propositional logic

- Proposition: declarative statements, either true or false but not both.

E.g. $1+1=2$ ✓
$1+3=7$ ✓
New Delhi is the capital of India. ✓
$5 > 10$ ✓

What is the time? ✗
$n+1=3$ ✗
Fuck off! ✗
The moon is beautiful. ✗

- We use letters to denote propositional variables (sentential variables).

e.g. p, q, r, a, b, c → atomic propositions

- Logical connectives: These symbols connect propositions to form compound statements.

- Negation (NOT):
Let P be a proposition. The negation of P is denoted as $\neg P$ or $\bar{P}$.
It is read as "It is not the case that P".

E.g. P: My PC runs Linux.
$\bar{P}$: It is not the case that my PC runs Linux / My PC does not run Linux.

Syntax: $\neg P$
Semantics: meaning of negation (truth table).

P	$\bar{P}$
T	F
F	T

Syntax, semantics

- Conjunction (AND):
Let P and Q be propositions. The conjunction of P and Q is denoted as $P \wedge Q$ and is read as "P and Q".

e.g. P: I went to the store. Q: I bought some apples.
$P \wedge Q$: I went to the store and bought some apples.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

- Disjunction OR (inclusive):

$P \vee Q$

True when atleast 1 of them is true.

P: you can go to a party.
Q: you can stay home and relax.
$P \vee Q$: you can go to the party or stay home and relax.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

- Exclusive OR:

$P \oplus Q$

↳ true when only one of them is true

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

P: A student can have salad with dinner.
Q: A student can have soup with dinner.

$P \oplus Q$: A student can have either salad or soup with dinner, but not both.

- Conditional statement (implication):

Let P and Q be propositions. The conditional statement "If P then Q" or "P implies Q" is denoted as $P \to Q$.

False when q is false even if p is true.

hypothesis / premise

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

e.g. p: It rains. q: we'll stay indoors. $p \to q$: If it rains, we'll stay indoors.

Different ways to express conditionals:

- if p then q $\equiv$ p implies q.
- if p, q $\equiv$ p only if q.
- p is sufficient for q $\equiv$ a sufficient condition for q is p.
- q if p $\equiv$ q whenever p
- q when p $\equiv$ q is necessary for p
- A necessary condition for p is q $\equiv$ q follows from p.

- Converse, inverse, and contrapositive

converse, inverse, contrapositive of $p \to q$

p	q	$\sim p$	$\sim q$	$p \to q$	$q \to p$	$\sim p \to \sim q$	$\sim q \to \sim p$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

$(p \to q) \equiv$ (contrapositive)

(converse) $\equiv$ (inverse)

p: It is raining q: The home team wins.

$p \to q$: If it is raining, the home team wins

$q \to p$: If the home team wins, it is raining. inverse

$\sim p \to \sim q$: If it is NOT raining, the home team doesn't win. converse

$\sim q \to \sim p$: If the home team doesn't win, then it isn't raining. contrapositive

- Biconditional (iff):

Let P and Q be propositions. The conditional statement "P iff Q" or "P if and only Q" is denoted as $P \Leftrightarrow Q$.
It is true when both P and Q have the same truth values.

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Different ways:

- iff / if and only if
- p is necessary and sufficient for q.
- If p then q and conversely. $(p \Leftrightarrow q) \equiv ((p \to q) \wedge (q \to p))$
- p exactly when q.

Propositional equivalences

- Tautology: A proposition which is always true irrespective of the truth values of the underlying variables.

e.g. $(p \vee \sim p)$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

- Contradiction: A proposition which is always false irrespective of the truth values of the underlying variables.

e.g. $(p \wedge \sim p)$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

- Contingency: A proposition which is neither a tautology nor a contradiction.

Q. $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$

→ tautology statement ≡ contrapositive

Q. $(p \rightarrow q) \rightarrow (p \wedge q)$

→

p	q	$p \wedge q$	$p \rightarrow q$	expression
T	T	T	T	T
T	F	F	F	T
F	T	F	T	F
F	F	F	T	F

Precedence of logical operators

$\sim$
$\wedge$
$\vee$
$\rightarrow$
$\leftrightarrow$

$(p \rightarrow q \wedge r) \equiv (p \rightarrow (q \wedge r))$

Equivalence

- $X \equiv Y$ iff they have the same truth values.
  (logically equivalent)
- If $X \equiv Y$, $X \leftrightarrow Y$ is a tautology.

Standard logically equivalent statements

- Identity laws

$p \wedge T \equiv p \qquad p \vee F \equiv p$

- Domination laws

$p \vee T \equiv T \qquad p \wedge F \equiv F$

- Idempotent laws

$p \vee p \equiv p \qquad p \wedge p \equiv p$

- Double negation law

$\sim(\sim p) \equiv p$

- Commutative laws

$p \wedge q \equiv q \wedge p \qquad p \vee q \equiv q \vee p$

- Associative laws

$(p \vee q) \vee (r) \equiv (p) \vee (q \vee r) \qquad (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

- Distributive laws

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r) \qquad p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

- Absorption laws

$p \vee (p \wedge q) \equiv p \qquad p \wedge (p \vee q) \equiv p$

- Negation laws

$p \wedge \sim p \equiv F \qquad p \wedge \sim p \equiv T$

- De Morgan's laws

$\sim(p \wedge q) \equiv \sim p \vee \sim q \qquad \sim(p \vee q) \equiv \sim p \wedge \sim q$

Logical equivalences involving conditionals

- $p \to q \equiv \sim p \vee q \equiv \sim q \to \sim p$
- $p \vee q \equiv \sim p \to q$
- $p \vee q \equiv \sim(p \to \sim q)$
- $\sim(p \to q) \equiv p \wedge \sim q$
- $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$
- $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
- $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$
- $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Logical equivalences involving bi-conditionals

- $(p \leftrightarrow q) \equiv (p \to q) \wedge (q \to p)$
- $(p \leftrightarrow q) \equiv (\sim p \leftrightarrow \sim q)$
- $(p \leftrightarrow q) \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$
- $\sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q$

Logical inference

Consistent systems:

- Given a set of logical statements, they should maintain consistency.
- These statements should NOT contain any contradictions.

E.g. A: It is sunny or it is warm. (p / q)
B: It is sunny
C: It is not warm.

$A \equiv p \vee q$
$B \equiv p$
$C \equiv \neg q$

cont. p	cont. $\neg q$	q	cont. $p \vee q$
T	T	F	T

A: If it is snowing (p), then it is cold (q).
B: It is snowing.
C: It is not cold.

$A: p \rightarrow q$
$B: p$
$C: \neg q$

p	$\neg q$	q	$p \rightarrow q$	A
T	T	F	F	T

$=$ inconsistent

(H.W.) Check the consistency of the following statements :-

① ✓
A: If it is raining, then the ground is wet.
B: It is raining or it is snowing.
C: The ground is not wet.

② ✓
A: If it is sunny, then it is warm.
B: It is sunny or it is snowing.
C: It is not warm.

→ p: It is sunny
q: It is warm
r: It is snowing

$A: p \rightarrow q$
$B: p \vee r$
$C: \neg q \Rightarrow q = \text{False}$

p	q	r	A: $\neg q$	B: $p \rightarrow q$	C: $p \vee r$
T	F	T	T	F	T
T	F	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	F

consistent

Arguments:

- A set of logical statements designed to establish the truth of a particular conclusion.

p_1
p_2
$\vdots$
p_n
$\therefore q$

premises
conclusion ← argument is valid if the premises lead to the conclusion

- An argument is valid, iff the conclusion logically follows the premises.
- The argument form is called valid, provided

$(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$

is true.

- Valid arguments can be sound or not.
Sound arguments: All of the premises are valid, not just some of them.

e.g. p_1: If you have the current password, then you can log onto the network.
p_2: You have the current password
$\therefore q$: You can log onto the network.

To check validity:

p: you have the current password
q: you can log onto the n/w

$p_1: p \rightarrow q$
$p_2: p$
$Q: q$

check $(p_1 \wedge p_2) \rightarrow Q$

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$((p \rightarrow q) \wedge p) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Tautology ⇒ The argument $(p_1 \wedge p_2) \models Q$ is valid

Q: check the validity:

P_1: If you have the current password, then you can log onto the network.

P_2: you can log onto the network.

Q: you have the current password.

→ $P_1: p \to q$ $\quad Q: p$

$P_2: q$

$(P_1 \wedge P_2)$

p	q	$p \to q$	$(p \to q) \wedge q$	$(p \to q) \wedge q \to p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

↳ invalid argument

Rules of inference

Modus ponens
(Rule of detachment)

$$P \to Q,\ P \vdash Q$$

Modus tollens
(Rule of contrapositives)

$$P \to Q,\ \neg Q \vdash \neg P$$

P	Q	$P \to Q$	$\neg Q$	$(P \to Q) \wedge (\neg Q)$	$\neg P$	$((P \to Q) \wedge (\neg Q)) \to (\neg P)$
T	T	T	F	F	F	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	T	T	T	T

Hypothetical syllogism
(transitivity)

$$\frac{P \to Q, Q \to R}{\therefore P \to R}$$

Disjunctive syllogism

$$\frac{P \vee Q,\ \neg P}{\therefore Q}$$

Conjunctive syllogism

$$\frac{P \vee Q,\ \neg Q}{\therefore P}$$

Addition

$$\frac{P}{\therefore P \vee Q}$$

Simplification

$$\frac{P \wedge Q}{\therefore P}$$

Conjunction

$$\frac{P, Q}{\therefore P \wedge Q}$$

Resolution

$$\frac{P \vee Q, \neg P \vee R}{\therefore Q \vee R}$$

Dilemma

$$\frac{P \vee Q,\ P \to R,\ Q \to R}{\therefore R}$$

Constructive dilemma

$$\frac{P \vee Q,\ P \to R,\ Q \to S}{\therefore R \vee S}$$

Destructive dilemma

$$\frac{P \to R,\ Q \to S,\ \neg R \to \neg S}{\therefore \neg P \vee \neg Q}$$

Q. Is the argument valid?
Statement: It is below freezing and raining now. Therefore, it is below freezing now.

simplification.

Q. $\dfrac{q}{p \to q}$ → valid

Q. $\dfrac{\sim(p \to q)}{p}$ → $\sim(\sim p \vee q) = p \wedge \sim q$ valid

Q. $\dfrac{\sim(p \to q)}{q}$ → $p \wedge \sim q$ invalid

Q. p_1: It is not sunny this afternoon ($\sim A$) and it is colder than yesterday (B).
p_2: we'll go swimming (C) only if it is sunny (A).
p_3: If we don't go swimming, we'll take a canoe trip (D).
p_4: If we take a trip (D), we'll be home by sunset (E).

$\sim A \wedge B$
$C \to A$
$\sim C \to D$
$D \to E$

conclusion: we'll be home by sunset (E).

→ A is false.

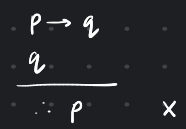
$(\sim C \vee A)$ ⇒ C = false
(T, F)

$\sim C \to D \Rightarrow D$

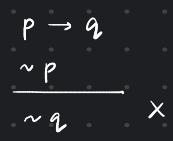
$D \to E \Rightarrow E$

Fallacies

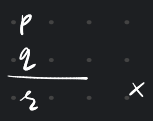
Affirming the consequence
(Fallacy of assuming converse)



Denying the antecedent
(Fallacy of assuming inverse)



Non sequitur
There's a gap in the logic that isn't explained.



> Life is life and fun is fun, but it's all so quiet when the goldfish die.
>
> — *West with the Night*, Beryl Markham[7]

Practice questions

① p : The election is decided.
q : The votes haven been counted.
Give the english sentence of these compound propositions.

a) $\neg q \rightarrow \neg p$

→ If the votes haven't been counted, then the election isn't decided.

b) $p \leftrightarrow q$

→ The election is decided iff the votes have been counted.

c) $\neg p \vee (\neg p \wedge q)$

→ $\sim p \vee (\sim p \wedge q)$
$= (\sim p \vee \sim p) \wedge (\sim p \vee q)$ } ×
$= (\sim p) \wedge (\sim p \vee q)$

Either the election is NOT decided
or else the election is not decided
and the votes have been counted.

⑪ p : You have the flu.
q : You miss the final examination.
r : You pass the course.
a) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
b) $(p \wedge q) \vee (\neg q \wedge r)$.

→ a) $(p \rightarrow \sim r) \vee (q \rightarrow \sim r)$
$\sim p \vee \sim r \vee \sim q \vee \sim r$ } ×
$\sim p \vee \sim q \vee \sim r$

NOTE: No reduction.

If you have the flu, then you won't pass the course or if you miss the final exam, you won't pass the course.

b) you'll have the flu and miss the exam or you won't miss the final exam and you'll pass.

② State the converse, contrapositive & inverse of each of them :-
a) I go to the beach whenever it is a sunny summer day.
b) When I stay up late, it is necessary the I sleep until noon.

$q \rightarrow p$ } converse
$\sim p \rightarrow \sim q$ } inverse
$\sim q \rightarrow \sim p$ } contrapositive

a) p : It is a sunny day q : I go to the beach.

conv. $q \rightarrow p$: If i go to the beach, it is a sunny day.

inv. $\sim p \rightarrow \sim q$: If it isn't a sunny day, then I don't go to the beach.

contra +ve $\sim q \rightarrow \sim p$: If I don't go to the beach, then it isn't a sunny day.

b) p : I stay up late q : It is necessary that I sleep until noon.

.
.

③ Which one of them is a tautology, contradiction & contingency.
a) $p \rightarrow \neg p$
b) $p \oplus (p \vee q)$
c) $p \leftrightarrow \neg p$

a) $p \rightarrow \sim p$
$\sim p \vee \sim p$
$\sim p$ contingency

b) $p \oplus (p \vee q)$

p	q	p∨q	p⊕(p∨q)
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

contingency

c) $p \leftrightarrow \neg p$

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

contradiction

d) $(p \wedge q) \rightarrow (p \vee q)$

e) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

f) $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

d) $\neg(p \wedge q) \vee p \vee q$

$\neg p \vee \neg q \vee p \vee q$

T

tautology

e)

p	q	$p \leftrightarrow q$	$q \rightarrow \neg p$	statement
T	T	T	F	F
T	F	F	T	F
F	T	F	T	F
F	F	T	T	T

contingency

f)

p	q	$p \leftrightarrow q$	$p \leftrightarrow \neg q$	statement
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	T	F	T

tautology

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1. Are these system specifications consistent?

A: Whenever the system software is being upgraded, users cannot access the file system.

B: If users can access the file system, then they can save new files.

C: If users cannot save new files, then the system software is not being upgraded.

p: sys. s/w is being upgraded

q: users cannot access the FS

r: users can save new files

A: $p \rightarrow q$

B: $\neg q \rightarrow r$

C: $\neg r \rightarrow \neg p$

$\neg r \rightarrow \neg p$

$r \vee \neg p$

$p \rightarrow r$

$p \rightarrow q$, $\neg q \rightarrow r$, $p \rightarrow r$ — put $r = q = T$ to make all 3 true

p	q	r	$p \rightarrow q$	$\neg q \rightarrow r$	$\neg r \rightarrow \neg p$
T	T	T	T	T	T
T	T	F	T	T	F
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	F	T

consistent

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p: FS is loaded

q: new messages will be queued

r: the system is functioning normally

s: msgs will be sent to msg buffer

A: $\neg p \rightarrow q$

B: $\neg p \leftrightarrow r$

C: $\neg q \rightarrow s$

D: $\neg p \rightarrow s$

E: $\neg s$

$s = F \Rightarrow p = q = F$

A: $T \rightarrow F$ ✗ F

inconsistent

③ Are they logically equivalent?

a) $(p \to q) \to r$ and $p \to (q \to r)$

b) $(p \wedge q) \to r$ and $(p \to r) \wedge (q \to r)$

→ a) $(p \to q) \to r$ $\quad$ $p \to (q \to r)$

$\sim(\sim p \vee q) \vee r$ $\quad$ $\sim p \vee (\sim q \vee r)$

$(p \wedge \sim q) \vee r$ $\quad$ $\sim p \vee \sim q \vee r$

$p = F$, $r = F$ } makes it False $\quad$ $p = F$, $r = F$ makes it true

b) $(p \wedge q) \to r$ $\quad$ $(p \to r) \wedge (q \to r)$

$\sim p \vee \sim q \vee r$ $\quad$ $(\sim p \vee r) \wedge (\sim q \vee r)$

$p = F$, $r = F$, $q = T$ } makes it true $\quad$ $p = F$, $r = F$, $q = T$ } makes it false

Dual: Any compound proposition that contains $\wedge$, $\vee$, and $\sim$ is turned to an expression with replacements:

$\wedge \to \vee$

$\vee \to \wedge$

$T \to F$

$F \to T$

④ Find the dual of each of these compound propositions

a) $p \vee \neg p$

b) $p \wedge (q \vee (r \wedge T))$

c) $(p \wedge \neg q) \vee (q \wedge F)$

$\sim$ is left untouched.

a) $S: p \vee \sim p$ $\quad$ b) $p \vee (q \wedge (r \vee F))$ $\quad$ c) $(p \vee \sim q) \wedge (q \vee F)$

$S^*: p \wedge \sim p$

(dual)

NOTE: $S^* \neq S$ $\quad$ S - compound proposition

$(S^*)^* = S$ $\quad$ S^* - its dual

⑤ Determine whether these arguments are valid or not.

a) Everyone enrolled in the university has lived in a dormitory. Mia has never lived in a dormitory. Therefore Mia is not enrolled in the university.

university → dorm $\equiv$ $\sim$univ $\vee$ dorm

$\sim$ dorm

$\sim$ university

valid

b) Quincy likes all action movies. Quincy likes the movie Eight Men Out. Therefore, Eight Men Out is an action movie.

action → likes

likes

action

invalid

c) A convertible car is fun to drive. Isaac's car is not a convertible. Therefore, Isaac's car is not fun to drive.

convertible → fun

$\sim$ convertible

$\sim$ fun

invalid

Determine whether these arguments are valid or not

(6) a) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.

x is +ve real $\rightarrow$ x^2 is +ve real

a^2 is +ve

$\therefore$ a is +ve real

invalid

b) If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.

valid

First-order logic (predicate calculus)

Statement: x is greater than 3 } not a proposition.

↳ can be denoted with a predicate function $P(x)$.

It becomes a proposition when x takes a concrete value. E.g. $P(3)$ is a proposition equal to False.

Quantifiers

- Used to convert predicates to propositions.
- Phrases like { all, some, many, none, few } are used in quantification.

Universal quantification (∀)

- The universal quantification of P(x) is a statement:
  P(x) for all values of x (in the domain).

$\forall x : P(x), \quad x \in \{x_1, \dots, x_n\}$

$P(x) = P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$ for finite domain

E.g. $P(x)$: $x^2 > 0$

If $\forall x\, P(x)$ true?

→ domain not specified.

True for $x \in \mathbb{R} - \{0\}$. False for $x \in \mathbb{R}$.

- A quantified statement is incomplete without domain specification.

Existential quantification (∃)

- The existential quantification of P(x) is a statement:
  P(x) for some values of x (in the domain).

$\exists x : P(x), \quad x \in \{x_1, x_2, \dots, x_n\}$

$P(x) = P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$

Uniqueness quantification (∃!)

- The uniqueness quantification of P(x) is a statement:
  P(x) such that x is unique.

$\exists! x$ or $\exists_{=1} x$

These quantifiers have a higher precendence than logical operators.

$\forall x\, P(x) \wedge Q(x) \equiv \left(\forall x\, P(x)\right) \wedge Q(x)$

$\not\equiv \forall x\, (P(x) \wedge Q(x))$

$\exists x : x + y = 1$

x: bounded variable; y: free variable

Free variable: lies outside the scope of all quantifiers.

Variables in diff. scopes can have same names.

$\forall x\, (P(x) \wedge Q(x)) \wedge \forall x\, R(x)$ is same as $\forall x\, (P(x) \wedge Q(x)) \wedge \forall y\, R(y)$

Logical equivalences involving predicates

Q. $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x)$ → True

$\forall x\,(P(x) \vee Q(x)) \equiv \forall x\,P(x) \vee \forall x\,Q(x)$ → False

De Morgan's laws:

$\sim \forall x\,P(x) \equiv \exists x \sim P(x)$

$\sim \exists x\,P(x) \equiv \forall x \sim P(x)$

Q. Write the negation of these statements :-

a) $\forall x\,(x^2 > x)$ — $\exists x\,(x^2 \leq x)$

b) $\exists x\,(x^2 = 2)$ — $\forall x\,(x^2 \neq 2)$

H.W

Q. Are they logically equivalent ?

a) $\neg \forall x\,(P(x) \rightarrow Q(x))$ & $\exists x\,(P(x) \wedge \neg Q(x))$

b) $\forall x\,(P(x) \rightarrow Q(x))$ & $\forall x\,P(x) \rightarrow \forall x\,Q(x)$

c) $\exists x\,P(x) \wedge \exists x\,Q(x)$ & $\exists x\,(P(x) \wedge Q(x))$.

d) $\forall x\,(P(x) \leftrightarrow Q(x))$ & $\forall x\,P(x) \leftrightarrow \forall x\,Q(x)$.

a) True $\exists x\,[P(x) \wedge \sim Q(x)]$ ✓✓

b) False $\forall x\,(\sim P(x) \vee Q(x))$ and $(\sim \forall x\,P(x) \vee \forall x\,Q(x))$ ✗✗

$P(x)$: x is divisible by 3.

$Q(x)$: x is divisible by 6.

c) False

c) $\exists x\,P(x) \wedge \exists x\,Q(x)$ & $\exists x\,(P(x) \wedge Q(x))$

$P(x)$ → all even numbers. $Q(x)$ → all odd numbers.

d) True

Inference rules for quantified statements

Universal instantiation

$$\frac{\forall x\ P(x)}{\therefore P(c)}$$

Universal generalization

$$\frac{\vdash P(x)}{\forall x\ P(x)}$$

Existential instantiation

$$\frac{\exists x\ P(x)}{P(c) \text{ for some element } c}$$

Existential generalization

$$\frac{P(c)}{\exists x\ P(x)}$$

Translations

- Every student in this class has studied calculus.

  $P(x)$: x is a student in this class

  $Q(x)$: x has studied calculus.

  Given statement: $\forall x\,(P(x) \rightarrow Q(x))$

Q. Translate these statements into English, domain → all animals.

$R(x)$: x is a rabbit

$H(x)$: x hops

a) $\forall x\,(R(x) \rightarrow H(x))$

b) $\forall x\,(R(x) \wedge H(x))$

c) $\exists x\,(R(x) \rightarrow H(x))$

d) $\exists x\,(R(x) \wedge H(x))$.

PYQs

3.1.15 First Order Logic: GATE CSE 2012 | Question: 13

What is the correct translation of the following statement into mathematical logic?

"Some real numbers are rational"

A. $\exists x(\text{real}(x) \vee \text{rational}(x))$
B. $\forall x(\text{real}(x) \rightarrow \text{rational}(x))$
C. $\exists x(\text{real}(x) \wedge \text{rational}(x))$ ✓
D. $\exists x(\text{rational}(x) \rightarrow \text{real}(x))$

gatecse-2012 mathematical-logic easy first-order-logic

3.1.17 First Order Logic: GATE CSE 2013 | Question: 47

Which one of the following is **NOT** logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

A. $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$ ✓
B. $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$ eq.
C. $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$ eq.
D. $\forall x(\exists y(\neg\alpha) \rightarrow \exists z(\neg\beta))$ ✓

mathematical-logic normal marks-to-all gatecse-2013 first-order-logic

$\forall x\left(\exists y(\bar{\alpha}) \vee \exists z(\bar{\beta})\right)$

$\left(\forall x\left[\forall y(\alpha) \rightarrow \exists z(\sim\beta)\right]\right.$

$\left.\forall x\left[\forall z(\beta) \rightarrow \exists y(\sim\alpha)\right]\right)$

3.1.13 First Order Logic: GATE CSE 2010 | Question: 30

Suppose the predicate $F(x, y, t)$ is used to represent the statement that person x can fool person y at time t.

Which one of the statements below expresses best the meaning of the formula,

$$\forall x \exists y \exists t(\neg F(x, y, t))$$

A. Everyone can fool some person at some time
B. No one can fool everyone all the time
C. Everyone cannot fool some person all the time
D. No one can fool some person at some time

$F(x,y,t) \rightarrow$ everyone can fool someone at some time t.

$\forall x\ \exists y\ \exists t\ (\sim F(x,y,t))$

$= \sim \exists x\ \forall y\ \forall t\ (F(x,y,t))$

$=$ no x exists such that x can fool all y's at all times t.

gatecse-2010 mathematical-logic easy first-order-logic

3.2.1 Logical Reasoning: GATE CSE 2012 | Question: 1

Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.

I_2: If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.

Which of the following is **TRUE**?

A. Both I_1 and I_2 are correct inferences
B. I_1 is correct but I_2 is not a correct inference ✓
C. I_1 is not correct but I_2 is a correct inference
D. Both I_1 and I_2 are not correct inferences

$\dfrac{r \rightarrow \sim c \quad ; \quad \sim c}{\sim r}$ $\quad = (\sim r \vee c)$ ✓

$(\sim c) \leftarrow c \equiv F$

$\dfrac{r \rightarrow \sim c \quad ; \quad \sim r}{c}$ ✗

3.3.25 Propositional Logic: GATE CSE 2008 | Question: 31

P and Q are two propositions. Which of the following logical expressions are equivalent?

I. $P \vee \neg Q$
II. $\neg(\neg P \wedge Q)$ $= P \vee \sim Q$
III. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$
IV. $(P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$

A. Only I and II
B. Only I, II and III ✓
C. Only I, II and IV
D. All of I, II, III and IV

gatecse-2008 normal mathematical-logic propositional-logic

a. $P + \bar{Q}$

b. $\overline{P\bar{Q}} = P + \bar{Q}$

c. $PQ + P\bar{Q} + \bar{P}\bar{Q}$
$P(Q+\bar{Q}) + \bar{P}\bar{Q}$
$P + \bar{P}\bar{Q}$
$(P+\bar{P})(P+\bar{Q})$
$= P + \bar{Q}$

d. $PQ + P\bar{Q} + \bar{P}Q$
$P(Q+\bar{Q}) + \bar{P}Q$
$P + \bar{P}Q$
$(P+\bar{P})(P+Q)$
$(P+Q)$

3.3.18 Propositional Logic: GATE CSE 2001 | Question: 1.3

Consider two well-formed formulas in propositional logic

$F_1 : P \Rightarrow \neg P \qquad F_2 : (P \Rightarrow \neg P) \vee (\neg P \Rightarrow P)$

Which one of the following statements is correct?

A. F_1 is satisfiable, F_2 is valid ✓
B. F_1 unsatisfiable, F_2 is satisfiable
C. F_1 is unsatisfiable, F_2 is valid
D. F_1 and F_2 are both satisfiable

gatecse-2001 mathematical-logic easy propositional-logic

$P \rightarrow \sim P$

$F \rightarrow T$ satisfiable
(true for atleast some combination of variables)

$(P \rightarrow \sim P) \vee (\sim P \rightarrow P)$

$(\bar{P} + \bar{P}) + (P + P)$

$\bar{P} + P$

T valid

3.3.15 Propositional Logic: GATE CSE 1998 | Question: 1.5

What is the converse of the following assertion?

- I stay only if you go

A. I stay if you go ✓
B. If I stay then you go
C. If you do not go then I do not stay
D. If I do not stay then you go

gate1998 mathematical-logic easy propositional-logic

$p \rightarrow q$ given: I stay $\rightarrow$ you go

converse: $q \rightarrow p$

If you go, I stay

3.3.22 Propositional Logic: GATE CSE 2004 | Question: 70

The following propositional statement is $(P \implies (Q \vee R)) \implies ((P \wedge Q) \implies R)$

A. satisfiable but not valid ✓
B. valid
C. a contradiction
D. None of the above

gatecse-2004 mathematical-logic normal propositional-logic

$$\overline{(\bar{P}+Q+R)} + (\overline{PQ}+R)$$
$$PQ\bar{R} + \bar{P} + \bar{Q} + R$$
$$\bar{Q}(1+P\bar{R}) + \bar{P} + R$$
$$\bar{Q} + \bar{P} + R$$

3.2.2 Logical Reasoning: GATE CSE 2015 Set 2 | Question: 3

Consider the following two statements.

- S_1: If a candidate is known to be corrupt, then he will not be elected
- S_2: If a candidate is kind, he will be elected

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

A. If a person is known to be corrupt, he is kind
B. If a person is not known to be corrupt, he is not kind
C. If a person is kind, he is not known to be corrupt
D. If a person is not kind, he is not known to be corrupt

gatecse-2015-set2 mathematical-logic normal logical-reasoning

$C \rightarrow \neg E$
$K \rightarrow E$

$\neg C \vee \neg E$
$E \rightarrow \neg C$

$K \rightarrow \neg C$

3.2.3 Logical Reasoning: GATE CSE 2015 Set 3 | Question: 24

In a room there are only two types of people, namely $\text{Type } 1$ and $\text{Type } 2$. $\text{Type } 1$ people always tell the truth and $\text{Type } 2$ people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking the person replies the following

"The result of the toss is head if and only if I am telling the truth"

Which of the following options is correct?

A. The result is head ✓
B. The result is tail
C. If the person is of $\text{Type } 2$, then the result is tail
D. If the person is of $\text{Type } 1$, then the result is tail

gatecse-2015-set3 mathematical-logic difficult logical-reasoning

head $\leftrightarrow$ speaks_truth

$P \leftrightarrow Q$

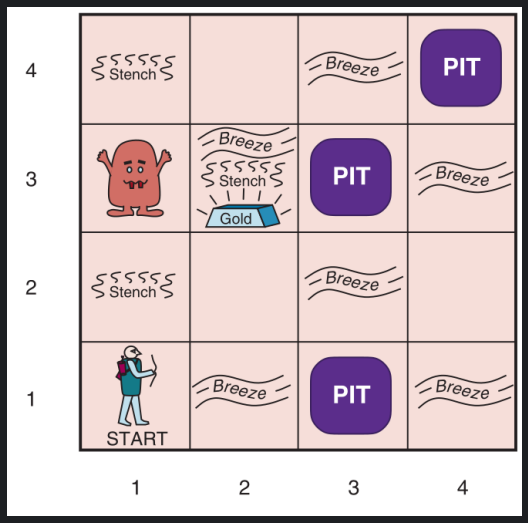
person can be type 1 / type 2

type 1 :	type 2 :
$(P \leftrightarrow Q)$ is true	$(P \leftrightarrow Q)$ is false
$P \leftrightarrow T$ is true	$P \leftrightarrow F$ is false
$\therefore P = T$	$P = T$

$\therefore$ head.

Wumpus world

- A classic problem in AI that demonstrates the use of knowledge representation and reasoning.



In this section we describe an environment in which knowledge-based agents can show their worth. The **wumpus world** is a cave consisting of rooms connected by passageways. Lurking somewhere in the cave is the terrible wumpus, a beast that eats anyone who enters its room. The wumpus can be shot by an agent, but the agent has only one arrow. Some rooms contain bottomless pits that will trap anyone who wanders into these rooms (except for the wumpus, which is too big to fall in). The only redeeming feature of this bleak environment is the possibility of finding a heap of gold. Although the wumpus world is rather tame by modern computer game standards, it illustrates some important points about intelligence.

- agent starts at (1, 1)
- actions → move foward
  - → turn left / right
  - → shoot an arrow to kill wumpus
  - → grab the gold
  - → climb outside the cave
- sensory info → breeze : felt near a pit
  - → stench : near the wumpus
  - → glitter : near the gold
  - → scream : heard when the wumpus is killed

We can encode rules like "if there is a breeze, a pit is nearby".

e.g. $\text{breeze}(1,1) \rightarrow (\text{pit}(1,2) \vee \text{pit}(2,1))$

actions like $(\text{wumpus_ahead} \wedge \text{wumpus_alive}) \rightarrow \text{shoot}$

$$\frac{(\text{wumpus_ahead} \wedge \text{wumpus_alive})}{\text{shoot}}$$

Horn clause

- A disjunction (OR) of literals with atmost 1 positive literal.

E.g. $(p \vee \neg q \vee \neg r)$ ✓

$(\neg p \vee \neg q)$ ✓

$(\neg p \vee q \vee r)$ ✗ not a horn clause

Definite horn clause: exactly one positive literal.
Negative horn clause: no positive literal.

Forward chaining

- Inference techniques, where reasoning starts from known facts and applies inference rules to derive new facts until the goal is reached or no further inference is possible.
- Data-driven approach.
- All possible conclusions are derived.
- E.g.

initial facts = {fever, cough, headache}

knowledge base = { fever ∧ cough → flu,
flu → take_rest,
headache ∧ fever → migraine }

pass 1: (fever ∧ cough) → flu
(headache ∧ fever) → migraine
} Inference → {flu, migraine}

pass 2: flu → take_rest } Inference → {take_rest}

pass 3: no new inference: termination.

Backward chaining

- Goal-driven approach.
- Starts with a goal, proves a specific conclusion.

e.g. KB: { (fever ∧ cough) → flu,
flu → take_rest,
(headache ∧ fever) → migraine
}

facts: {fever, cough, headache}
goal: prove take_rest

→ say take_rest = True
pass 1: prove take_rest
pass 2: prove flu because flu → take_rest

SAT problem

- A compound proposition is satisfiable, if true for atleast 1 case.

e.g. $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ → product of sums (conjunctive normal form)

- The SAT problem is to check if a formula x is satisfiable or not.

Q. CNF or not?

a) $p \vee q$ → yes $(p \vee q) \wedge T$

b) p → yes $(p) \wedge T$

Q.29	Let x and y be two propositions. Which of the following statements **is a tautology /are tautologies**?
(A)	$(\neg x \wedge y) \Longrightarrow (y \Longrightarrow x)$
(B)	$(x \wedge \neg y) \Longrightarrow (\neg x \Longrightarrow y)$ ✓
(C)	$(\neg x \wedge y) \Longrightarrow (\neg x \Longrightarrow y)$ ✓
(D)	$(x \wedge \neg y) \Longrightarrow (y \Longrightarrow x)$ ✓

a) $\bar{x}y \to (y \to x)$

$\bar{x}\bar{y} + \bar{y} + x$

$x + \bar{y} + \bar{y} + x$

$x + \bar{y}$

b) $x\bar{y} \to (\bar{x} \to y)$

$\bar{x} + y + x + y$

$T + y$

T

c) $\bar{x}y \to (\bar{x} \to y)$

$x + \bar{y} + x + y$

$x + T$

T

d) $x\bar{y} \to (y \to x)$

$\bar{x} + y + \bar{y} + x$

T

Q.54	Let *game(ball, rugby)* be true if the ball is used in rugby and false otherwise. Let *shape(ball, round)* be true if the ball is round and false otherwise. Consider the following logical sentences: *s1:* $\forall ball \neg\, game(ball, rugby) \Longrightarrow shape(ball, round)$ *s2:* $\forall ball \neg\, shape(ball, round) \Longrightarrow game(ball, rugby)$ *s3:* $\forall ball\ game(ball, rugby) \Longrightarrow \neg\, shape(ball, round)$ *s4:* $\forall ball\ shape(ball, round) \Longrightarrow \neg\, game(ball, rugby)$ Which of the following choices is/are logical representations of the assertion, "All balls are round except balls used in rugby"?
(A)	$s1 \wedge s3$ ✓
(B)	$s1 \wedge s2$
(C)	$s2 \wedge s3$ ✓
(D)	$s3 \wedge s4$

} all non-rugby balls are round

} all non-round balls are used in rugby

} all rugby balls are non-round

} all round balls are non-rugby balls

S_1: rugby ∨ round

S_2: round ∨ rugby

S_3: $\overline{\text{rugby}}$ ∨ $\overline{\text{round}}$

S_4: $\overline{\text{round}}$ ∨ rugby

$S_1 \equiv S_2$

$S_3 \equiv S_4$

statement: $\forall$ balls (round if rugby) ∧ ($\overline{\text{round}}$ if rugby)

$\equiv$ (rugby ∨ round) ∧ ($\overline{\text{round}}$ ∨ $\overline{\text{rugby}}$)

$\equiv S_1 \wedge S_3 \equiv S_2 \wedge S_3 \equiv S_1 \wedge S_4 \equiv S_2 \wedge S_4$

same as round ⟷ $\overline{\text{rugby}}$

Q.6	Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk. Based on the given passage, which of the following can be inferred?
(A)	All human beings can digest dairy milk. some ✗
(B)	No human being can digest dairy milk. ✗
(C)	Digestion of dairy milk is essential for human beings. ✗
(D)	In human beings, digestion of dairy milk resulted from a mutated gene. ✓

dairy ∧ mutn to digest in some people

GATE 2024